Volume IV · Chapter 18 — Ingestion, OCR & Document Understanding · Theory

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-IV-Chapter-18-context.md` for the AI interaction log.

Prerequisites: Ch 5, Ch 12.

Learning outcomes — the student can: parse born-digital and scanned medical books; run OCR handling medical vocabulary; extract layout/structure (sections, tables, references); build a clean, provenance-tagged text corpus.

Topics (theory) — 8 h:

#	Topic	h
18.1	Document types & challenges: PDFs, scans, layout, columns	2
18.2	OCR fundamentals & medical-text pitfalls (terms, dosages, tables)	2
18.3	Layout analysis & structure extraction (sections, headings, refs)	2
18.4	Tables, equations & footnotes; reading order	1
18.5	Provenance & metadata capture during ingestion	1

Datasets/tools: licensed/open medical texts; PyMuPDF, Tesseract, unstructured/Nougat/Marker. **Assessment:** ingestion pipeline + corpus (**60%**); OCR-quality report (**20%**); quiz (**20%**). **Key decisions:** born-digital vs. scanned handling; OCR engine; how much structure to preserve; **licensing (owned/licensed sources only)**. **References:** plan §6; §13. **Hours:** Theory **8** + Lab **14** = **22**.